NLP Techniques - Unit 2

Code Reference

See Unit 2.0.ipynb for implementation examples of:

- **POS (Part of Speech) Tagging** - Identifying grammatical roles of words
- **Named Entity Recognition (NER)** - Extracting named entities from text
- **Language Model Vectorization** - Converting language models to vector representations
- **Word Embedding** - Mapping words to dense vector spaces

One Hot Encoding

One hot encoding is a technique used to convert categorical data into a binary format. Each category is represented as a binary vector where only one element is "hot" (1) and the rest are "cold" (0). This is commonly used in NLP for representing words or tokens in a way that can be processed by machine learning algorithms.

Example

Documents:

- Document #1: "He is a good boy. She is also good."
- Document #2: "Radhika is a good person."

Vocabulary: a, also, boy, good, He, is, person, She, Radhika

Word Index Mapping:

Word	Index
a	0
also	1
boy	2
good	3
He	4
is	5
person	6
She	7
Radhika	8

One Hot Encoding Representation:

Each word is represented as a vector where only the position corresponding to its index is 1, and all other positions are 0.

Word	One Hot Encoding
a	[1, 0, 0, 0, 0, 0, 0, 0, 0]
also	[0, 1, 0, 0, 0, 0, 0, 0, 0]
boy	[0, 0, 1, 0, 0, 0, 0, 0, 0]
good	[0, 0, 0, 1, 0, 0, 0, 0, 0]
He	[0, 0, 0, 0, 1, 0, 0, 0, 0]
is	[0, 0, 0, 0, 0, 1, 0, 0, 0]
person	[0, 0, 0, 0, 0, 0, 1, 0, 0]
She	[0, 0, 0, 0, 0, 0, 0, 1, 0]
Radhika	[0, 0, 0, 0, 0, 0, 0, 0, 1]

Document as Matrix

When we apply one-hot encoding to all words in a document, we create a matrix where each row represents a word's one-hot encoding. The words appear in the order they occur in the document.

Document #1: "He is a good boy. She is also good."

Word	Index	0	1	2	3	4	5	6	7	8
He	4	0	0	0	0	**1**	0	0	0	0
is	5	0	0	0	0	0	**1**	0	0	0
a	0	**1**	0	0	0	0	0	0	0	0
good	3	0	0	0	**1**	0	0	0	0	0
boy	2	0	0	**1**	0	0	0	0	0	0
She	7	0	0	0	0	0	0	0	**1**	0
is	5	0	0	0	0	0	**1**	0	0	0
also	1	0	**1**	0	0	0	0	0	0	0
good	3	0	0	0	**1**	0	0	0	0	0

Document #1 → 9 × 9 matrix

Document #2: "Radhika is a good person."

Word	Index	0	1	2	3	4	5	6	7	8
Radhika	8	0	0	0	0	0	0	0	0	**1**

Word	Index	0	1	2	3	4	5	6	7	8
is	5	0	0	0	0	0	**1**	0	0	0
a	0	**1**	0	0	0	0	0	0	0	0
good	3	0	0	0	**1**	0	0	0	0	0
person	6	0	0	0	0	0	0	**1**	0	0

Document #2 → 5 × 9 matrix

Each document becomes a 2D matrix where:

- **Rows** = number of words in the document
- **Columns** = size of the vocabulary
- Each row contains exactly one "1" (at the index corresponding to that word) and the rest are "0"s

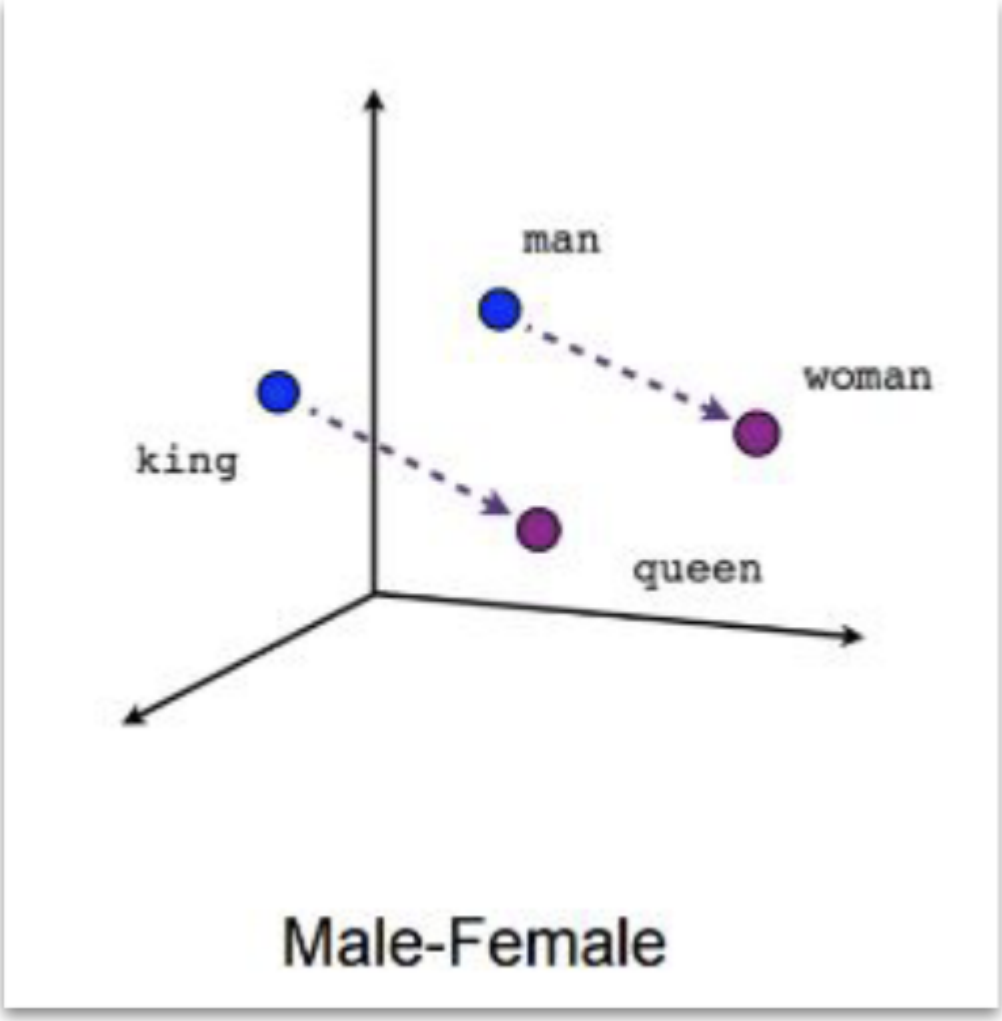
Issues with One Hot Encoding

1. **Curse of Dimensionality** - For large vocabulary sizes (common in NLP problems), the document matrix becomes huge and sparse, leading to increased computational and memory overhead.

2. **Loss of Semantic Information** - One hot encoding vectors do not provide any information about how words are related to each other. There is no semantic relationship captured between similar or related words.

Word2Vec

Word2Vec is a popular word embedding technique that converts words into dense vector representations. It captures semantic relationships between words by learning an embedding space where meaningful relationships often appear as consistent **directions**.



Word2Vec uses two main architectures:

- **CBOW (Continuous Bag of Words)**: predicts the target word from surrounding context.
- **Skip-gram**: predicts surrounding context words from a target word.

Context Window (Window Size)

Word2Vec learns embeddings from **neighbouring words**. A *window size* w means we consider up to w words to the **left** and w words to the **right** of a target word as its context.

Sentence: "The Sun rises in the east"

Diagram (target word = `in`)

```
flowchart LR
  The[The] --- Sun[Sun] --- rises[rises] --- IN[in] --- the[the] ---
east[east]

  classDef target fill:#FFF2CC,stroke:#B45F06,stroke-width:2px;
  class IN target;
```

Window size = 1

```
flowchart LR
  rises[rises] --> IN[in]
  the[the] --> IN[in]

  classDef target fill:#FFF2CC,stroke:#B45F06,stroke-width:2px;
  classDef ctx fill:#EAD1DC,stroke:#741B47;
  class IN target;
  class rises,the ctx;
```

Context, target pairs (like the slide notation): **(rises, in)**, **(the, in)**

For each target word, take 1 word to the left and 1 word to the right (when available):

Target	Context words
The	Sun
Sun	The, rises
rises	Sun, in
in	rises, the
the	in, east
east	the

Example (Skip-gram style) **(target → context)** pairs:

- (Sun → The), (Sun → rises)
- (in → rises), (in → the)

Window size = 2

```
flowchart LR
  Sun[Sun] --> IN[in]
  rises[rises] --> IN[in]
  the[the] --> IN[in]
  east[east] --> IN[in]

  classDef target fill:#FFF2CC,stroke:#B45F06,stroke-width:2px;
  classDef ctx fill:#EAD1DC,stroke:#741B47;
  class IN target;
  class Sun,rises,the,east ctx;
```

Context, target pairs: **(Sun, in)**, **(rises, in)**, **(the, in)**, **(east, in)**

For each target word, take up to 2 words on each side:

Target	Context words

Target	Context words
The	Sun, rises
Sun	The, rises, in
rises	The, Sun, in, the
in	Sun, rises, the, east
the	rises, in, east
east	in, the

This is how Word2Vec creates training examples in an **unsupervised** way: by predicting a word from its neighbours (CBOW) or predicting neighbours from a word (Skip-gram).

Vector Relationships (Analogies)

In Word2Vec embeddings, some relationships can be expressed using vector arithmetic (analogy-style reasoning):

- **Male–Female**: king – man + woman ≈ queen
- **Verb tense**: walked – walking ≈ swam – swimming
- **Country–Capital**: Italy – Rome ≈ Germany – Berlin (and similarly for other country–capital pairs)

CBOW (Continuous Bag of Words)

In **CBOW**, the model takes a set of **context words** (neighbours) as input and tries to predict the **target word** in the middle.

Neural-network view (Input → Hidden → Output)

- **Input layer**: the context words (often represented as one-hot vectors or word IDs).
- **Hidden layer**: the "Neural Network" box in the slide. In classic Word2Vec, this is effectively the **embedding lookup + averaging/summing** of the context word embeddings (a lightweight network, not a deep model).
- **Output (target) layer**: a probability distribution over the vocabulary (typically via softmax / negative sampling) to predict the **target word**.

Example (window size = 2):

Sentence: "The Sun rises in the east"

Context words: **The, Sun, in, the** → Target word: **rises**

```
flowchart LR
  subgraph C[Context words]
    w1[The]
    w2[Sun]
    w3[in]
    w4[the]
```

```
    end

    h["Hidden layer (embedding lookup, avg sum)"]
    o["Output layer (softmax over vocab)"]
    t["rises (target word)"]

    w1 --> h
    w2 --> h
    w3 --> h
    w4 --> h
    h --> o --> t
```

Key idea: CBOW learns word embeddings by repeatedly answering: **"Given neighbouring words, what is the most likely middle word?"**

Skip-gram

In **Skip-gram**, the model does the opposite of CBOW: it takes a **target word** as input and tries to predict the **context words** around it.

Neural-network view (Input → Hidden → Output)

- **Input layer**: the target word (one-hot vector or word ID).
- **Hidden layer**: the "Neural Network" box (embedding lookup for the target word).
- **Output layer**: predicts one or more nearby context words (again via softmax / negative sampling).

Example (window size = 2):

Sentence: "The Sun rises in the east"

Target word: **rises** → Context words: **The, Sun, in, the**

```
  flowchart LR
    x["rises (target word)"] --> h["Hidden layer (target embedding)"]
    h --> o1[The]
    h --> o2[Sun]
    h --> o3[in]
    h --> o4[the]
```

Key idea: Skip-gram learns embeddings by answering: **"Given this word, which words tend to appear near it?"**

Word2Vec Neural Network Architecture

Word2Vec uses a **simple shallow neural network** — just three layers:

```
Input Layer  →  Hidden Layer  →  Output Layer
(one-hot)       (embeddings)      (softmax)
```

Input Layer

Each word is fed into the network as a **one-hot vector**.

- **Size of the input vector = vocabulary size**
- If we have a vocabulary of **10,000 words**, the input vector has **10,000 dimensions**
- All values are `0` except one `1` at the index of the target word

Example — the word "king" with a 10,000-word vocabulary:

```
Index:   0    1    2    ...   [4500]   ...   9999
Vector: [0,   0,   0,   ...,    1,    ...,    0 ]
                                ↑
                          "king" = index 4500
```

Property	Value
Vector type	One-hot
Vector length	Vocabulary size (e.g. 10,000)
Non-zero values	Exactly 1
Captures semantics?	No — all words are equally distant

Hidden Layer

The hidden layer is the heart of Word2Vec — this is where the **embeddings live**.

- You choose the number of neurons (a hyperparameter)
- A common choice is **50** neurons (or 100, 200, 300 — your choice)
- The hidden layer has **no activation function** (linear projection)
- Each word's hidden layer output = its **word embedding vector**

```
Input layer        Hidden layer
(10,000 units)     (50 units)

  [ 0 ]               ( o )
  [ 0 ]          /    ( o )
  [ 0 ]    W1   /     ( o )
  [ 1 ] ————————  →   ( o )    ← 50 numbers for "king"
  [ 0 ]          \    ( o )
  [ 0 ]           \   ( o )
  [ 0 ]
```

```
  ...
(10,000 dims)     (50 dims)
```

Property	Value
Number of neurons	50 (hyperparameter — choose freely)
Output per word	50 numbers (the word embedding)
Activation	None (linear)
Weight matrix W_1	Shape: 10,000 × 50

> The weight matrix $\mathbf{W_1}$ is what gets trained. After training, each **row** of W_1 is the embedding for one word.

Output Layer

The output layer reconstructs the vocabulary distribution to make predictions.

- **Size = vocabulary size** (10,000)
- Uses **softmax** to output a probability over all words
- During training: predicts context words (Skip-gram) or target word (CBOW)

```
Hidden layer       Output layer
(50 units)         (10,000 units)

  ( o )                [ ]
  ( o )   W2    /      [ ]
  ( o ) ──────────►    [ ]      10,000 predicted probabilities
  ( o )         \     [ ]
  ( o )                [ ]
                       ...
  (50 dims)        (10,000 dims)
```

Property	Value
Number of outputs	Same as vocabulary size (10,000)
Activation	Softmax → probability distribution
Weight matrix W_2	Shape: 50 × 10,000

> **Problem**: 10,000 softmax outputs is very expensive. **Solution**: Use **Negative Sampling** or **Hierarchical Softmax** to approximate it efficiently.

Getting the Word Embeddings

After training is complete, **the output layer is discarded entirely** — it was only needed during training.

```
Input layer        Hidden layer        Output layer
(10,000)           (50)                (10,000)
                                           ×  ← discarded after training
    [one-hot]  →  [embedding]  →  [softmax]
```

The **output of the hidden layer IS the word embedding**:

- For each of the 10,000 words in the vocabulary, the hidden layer produces **50 numbers**
- These 50 numbers are the word's **embedding vector** — a dense, meaningful representation
- **Embedding size = number of neurons in the hidden layer = 50**

The "king" Embedding — End to End

Using our example (vocab = 10,000, hidden = 50):

Step 1 — One-hot input

```
"king"  →  [0, 0, 0, ..., 1, ..., 0]   (10,000-dim, 1 at index 4500)
```

Step 2 — Multiply by W_1 (10,000 × 50)

```
one-hot × W₁  =  row 4500 of W₁  =  king's embedding
```

Because it's a one-hot vector, this is just a **lookup** of row 4500 — no real multiplication needed.

Step 3 — Hidden layer output (the embedding)

```
king  →  [ 0.71,  -0.23,  0.88,  0.14,  -0.55,  0.32,  ...,  0.19 ]
                                                                (50
numbers)
```

Step 4 — Discard output layer, keep the 50 numbers

```
king_vector = [ 0.71, -0.23, 0.88, 0.14, -0.55, 0.32, ..., 0.19 ]
                └──────────────── 50 dimensions ────────────────┘
```

This dense 50-number vector is what gets stored and used for downstream NLP tasks.

Summary: Full Architecture

Layer	Size	Role
Input	10,000 (vocab size)	One-hot encoding of the word
Hidden	50 (chosen)	Word embedding — the output we want
Output	10,000 (vocab size)	Predicts context/target — discarded after training

```
                    Training                    Inference
                    ────────                    ─────────
"king"              one-hot       embedding     output
(word)   ──────▶  [0..1..0]  →  [50 nums]  →  [10,000]
                                     ↑              ↑
                                 KEEP THIS      THROW AWAY
```

> **Key insight**: Word2Vec doesn't care about the prediction task — the prediction is just a training signal. What we actually want is the **hidden layer weights** (W_1), which encode the semantic meaning of every word in just 50 numbers.

FastText (by Facebook Research)

FastText extends Word2Vec by operating at the **subword level** instead of only the word level.

Key Ideas

1. Very similar to Word2Vec in looking at **neighbours of a word**
2. Word embedding not only at **word level** but at **subword level**
3. Subwords are created as **character n-grams**
4. Can handle **unseen or rare words**

What is Subword Level?

Instead of treating "Facebook" as one atomic token, FastText breaks it into character n-grams:

```
"Facebook"  →  "Face book"  →  fa, ce, bo, ok   (subword level)
```

Character N-gram Example

Word: `awesome` (with n-gram size 3)

```
awesome  →  awe, wes, eso, som, ome
```

Each subword gets its own embedding. The final word embedding = **sum of all its subword embeddings**.

Subword	Embedding

Subword	Embedding
`awe`	[0.2, -0.1, ...]
`wes`	[0.5, 0.3, ...]
`eso`	[-0.1, 0.4, ...]
`som`	[0.3, -0.2, ...]
`ome`	[0.1, 0.5, ...]
awesome	**sum of all above**

Why Subwords Help

Scenario	Word2Vec	FastText
Word in training vocab	✓ Has embedding	✓ Has embedding
Unseen word (OOV)	✕ No embedding	✓ Built from subwords
Rare word (few examples)	Poor embedding	Better — shares subwords with common words
Morphologically rich languages	Struggles	Handles well (prefixes, suffixes)

> **Example**: FastText can embed `"Facebooking"` even if it never appeared in training, because it recognises `"face"`, `"book"`, `"ing"` as known subwords.

Global Vectors — GloVe (Stanford NLP)

GloVe is a word embedding technique developed by **Stanford University (2014)**.

Key Ideas

1. Similar to Word2Vec in looking at **neighbours of a word**
2. But also takes into account **how many times two words were neighbours** (global co-occurrence counts)
3. Approach provided by **Stanford University (2014)**
4. Lots of **pre-trained models** (with different embedding sizes) available

How GloVe Differs from Word2Vec

Word2Vec learns from a **sliding window** (local context only). GloVe builds a **global co-occurrence matrix** first — counting how often every word pair appears across the entire corpus — then factorises it into embeddings.

```
Word2Vec:  learn from local windows one sentence at a time
GloVe:     build global count matrix → factorise → embeddings
```

Co-occurrence matrix example:

	ice	steam	water	cold	hot
ice	—	0.8	1.5	3.2	0.1
steam	0.8	—	1.2	0.1	3.1

The ratio `P(cold|ice) / P(cold|steam)` is much greater than 1 → GloVe captures that *cold is related to ice*. This **ratio of co-occurrence probabilities** is what GloVe optimises — making it more stable than Word2Vec on large corpora.

Pre-trained GloVe Models

Model	Corpus	Vocab	Dims
`glove.6B`	Wikipedia + Gigaword	400K	50 / 100 / 200 / 300
`glove.42B`	Common Crawl	1.9M	300
`glove.840B`	Common Crawl	2.2M	300
`glove.twitter.27B`	Twitter	1.2M	25 / 50 / 100 / 200

Word2Vec vs FastText vs GloVe — Comparison

Feature	Word2Vec	FastText	GloVe
Developer	Google (2013)	Facebook Research (2017)	Stanford NLP (2014)
Training approach	Local sliding window (CBOW / Skip-gram)	Local sliding window + subword n-grams	Global co-occurrence matrix factorisation
Unit of representation	Whole word	Word + character n-grams	Whole word
Handles OOV words?	✕ No	✓ Yes — via subwords	✕ No
Handles rare words?	Poor	Good — shares subword info	Moderate
Uses global statistics?	✕ No	✕ No	✓ Yes
Training speed	Fast	Slightly slower (subwords)	Moderate (matrix build)
Pre-trained models?	✓ Google News (3M words, 300d)	✓ 157 languages	✓ Multiple corpora & sizes
Morphologically rich languages	Poor	Best	Moderate
Embedding type	Dense static	Dense static	Dense static

Feature	Word2Vec	FastText	GloVe
Context-aware?	✕ One vector per word	✕ One vector per word	✕ One vector per word
Best for	General NLP baseline	Low-resource / rare-word tasks	Large corpora, stable training

When to Use Which

Scenario	Recommendation
General English NLP task	**GloVe** (stable, good pre-trained models)
Many rare words or OOV expected	**FastText**
Non-English or morphologically rich language	**FastText**
Need interpretable, fast baseline	**Word2Vec**
State-of-the-art accuracy needed	**BERT / contextual embeddings** (all three above are static)

> **All three are static embeddings** — bank gets the same vector whether it means a river bank or a financial bank. Contextual models (BERT, GPT) solve this by generating a different vector for each occurrence depending on context.

Using a Pre-trained Word2Vec Model

Instead of training from scratch, we can reuse embeddings already trained on massive corpora. Pre-trained models are available at: github.com/RaRe-Technologies/gensim-data

Why Pre-trained?

Train from scratch	Use pre-trained
Needs large corpus	Works immediately
Time and compute intensive	Load and use in seconds
Embeddings reflect your domain	Embeddings reflect general language
Best for very domain-specific text	Best for general NLP tasks

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Document Vectorization — End to End Example

To feed text into an ML model, every document must become a **fixed-length numerical vector**. Below is the complete pipeline using two example documents.

Step 1 — The Documents

	Content
Document #1	"He is a good boy. She is also good."
Document #2	"Radhika is a good person."

Step 2 — Build Vocabulary & Assign Index

Collect all unique words across all documents, then assign each a numeric index.

Vocabulary:

Word	a	also	boy	good	He	is	person	She	Radhika
Index	1	2	3	4	5	6	7	8	9

> Each unique word gets exactly one index. Index `0` is reserved for **padding** (dummy words).

Step 3 — Create Document Vector (Replace each word by its index)

Map every word in each document to its vocabulary index, in order:

Document #1: `"He is a good boy. She is also good."`

```
He   →  5
is   →  6
a    →  1
good→  4
boy →  3
She →  8
is   →  6
also→  2
good→  4

Document #1 vector: [5, 6, 1, 4, 3, 8, 6, 2, 4]   (length = 9)
```

Document #2: `"Radhika is a good person."`

```
Radhika →  9
is      →  6
a       →  1
good    →  4
person  →  7

Document #2 vector: [9, 6, 1, 4, 7]   (length = 5)
```

Document	Vector	Length

Document	Vector	Length
Doc #1	`[5, 6, 1, 4, 3, 8, 6, 2, 4]`	9
Doc #2	`[9, 6, 1, 4, 7]`	5

Step 4 — Make Document Vectors the Same Length

ML models require all input vectors to be **the same fixed length**.

Rules:

1. Decide on a **chosen sequence length** (e.g. **8 words**)
2. If a document has **more** words than the chosen size → **truncate** (drop extra words)
3. If a document has **fewer** words than the chosen size → **pad** with dummy word index `0`

Chosen length = 8

Document #1 — Truncation (length 9 → 8, drop the last word):

```
Before: [5, 6, 1, 4, 3, 8, 6, 2, 4]   ← length 9
After:  [5, 6, 1, 4, 3, 8, 6, 2]       ← length 8  (last element dropped)
```

Document #2 — Padding (length 5 → 8, add three `0`s):

```
Before: [9, 6, 1, 4, 7]                    ← length 5
After:  [9, 6, 1, 4, 7, 0, 0, 0]           ← length 8  (padded with 0s)
```

Document	After Truncation / Padding	Action
Doc #1	`[5, 6, 1, 4, 3, 8, 6, 2]`	Truncated (9 → 8)
Doc #2	`[9, 6, 1, 4, 7, 0, 0, 0]`	Padded (5 → 8)

Both documents are now **length 8** — ready for an ML model.

Full Pipeline Summary

```
Raw text
   │
   ├─ 1. Tokenize each document
   ├─ 2. Build vocabulary → assign integer index to each word
   ├─ 3. Convert each document to index sequence
   ├─ 4. Decide fixed sequence length L
   │       ├─ len > L  →  truncate (keep first L tokens)
   │       └─ len < L  →  pad with 0s at the end
```

```
    └─ 5. Feed fixed-length integer sequences into the model
              (Embedding layer maps each index to its word vector)
```

> **In Keras / PyTorch**, this is handled by:
>
> - `Tokenizer` + `pad_sequences` (Keras)
> - `torchtext` vocabulary + `nn.utils.rnn.pad_sequence` (PyTorch)
> - The `Embedding` layer then looks up the pre-trained Word2Vec / GloVe vector for each index.

Text Similarity

Text Similarity measures *how close two chunks of text are by meaning*.

> It is not just about matching exact words — two sentences can use completely different words yet express the same idea. True text similarity captures **semantic closeness**, not just surface overlap.

Where Text Similarity is Useful

Application	How Text Similarity Helps
Web Search	Compare a search query against document text to rank results by relevance
Chatbots	Match a user's question to a known FAQ answer
Recommendation	Find products, articles, or movies with similar descriptions
Plagiarism Detection	Identify documents with the same meaning even if reworded
Duplicate Detection	Merge records that refer to the same entity
Machine Translation QA	Check if a translated sentence preserves the original meaning

---

How to Check Text Similarity

The process has two stages:

```
Text #1 ─┐                          ┌─► Similarity Score
         ├─►  Build Features ─►  Compare Features
Text #2 ─┘                          └─► (0.0 = unrelated, 1.0 =
identical)
```

Stage 1 — Build Features

Convert each text into a numerical representation using one of:

Method	Type	Captures Meaning?
Count (BoW)	Sparse vector	No — word frequency only

Method	Type	Captures Meaning?
TF-IDF	Sparse vector	Partial — weights rare words
Word2Vec	Dense vector	Yes — semantic relationships
GloVe	Dense vector	Yes — global co-occurrence
FastText	Dense vector	Yes — subword aware

Stage 2 — Compare Features

Once both texts are vectors, compute a similarity (or distance) score:

Metric	Formula	Range	Meaning
Cosine Similarity	`cos(θ) = (A · B) / (‖A‖ ‖B‖)`	0 → 1	1 = same direction = same meaning
Euclidean Distance	$d = \sqrt{\Sigma(a_i - b_i)^2}$	0 → ∞	0 = identical; larger = more different
Jaccard Index	`|A ∩ B| / |A ∪ B|`	0 → 1	1 = same words; 0 = no overlap
ML Model	Learned function	0 → 1	Trained end-to-end (e.g. BERT + cosine)

Cosine Similarity — Deep Dive

Cosine similarity is the **most widely used** metric for text similarity because it is **angle-based**, not magnitude-based — a short document and a long document about the same topic will still score highly.

```
          B
         /
        /  θ (small angle = similar)
       /____
      A

cos(θ) close to 1  →  texts are similar
cos(θ) close to 0  →  texts are unrelated
```

Formula:

$$\cos(\theta) = \frac{A \cdot B}{\|A\| \; \|B\|} = \frac{\Sigma(a_i \times b_i)}{\sqrt{\Sigma a_i^2} \times \sqrt{\Sigma b_i^2}}$$

Example with TF-IDF vectors:

	"I love NLP"	"I enjoy NLP"	"The stock fell"
"I love NLP"	1.00	~0.85	~0.05
"I enjoy NLP"	~0.85	1.00	~0.05
"The stock fell"	~0.05	~0.05	1.00

Jaccard Index — Deep Dive

Jaccard measures **word-set overlap** — useful for short texts, keywords, or tag matching.

```
         |A ∩ B|         words in BOTH texts
J(A,B) = ———————  =  ——————————————————————————
         |A ∪ B|      words in EITHER text (union)
```

Example:

```
Text A: {"the", "cat", "sat", "on", "mat"}
Text B: {"the", "cat", "sat", "on", "log"}

Intersection: {"the", "cat", "sat", "on"}  → 4 words
Union:        {"the", "cat", "sat", "on", "mat", "log"}  → 6 words

J(A, B) = 4 / 6 = 0.67
```

> Jaccard ignores word frequency and order — it is best for **keyword matching and set-based comparisons**, not semantic meaning.

Euclidean Distance — Deep Dive

Euclidean distance measures the **straight-line distance** between two vectors in embedding space.

```
d(A, B) = √[ (a₁−b₁)² + (a₂−b₂)² + ... + (aₙ−bₙ)² ]
```

- **d = 0** → identical vectors
- **d large** → vectors far apart = texts are different

> **Limitation**: Euclidean distance is sensitive to vector magnitude (length of document). A long document can appear distant from a short one even if they cover the same topic. This is why **cosine similarity is preferred** for text.

Choosing the Right Approach

```
Build Features          Compare Features       Best For
────────────────        ────────────────       ────────
Count / TF-IDF    ──▶   Cosine Similarity  ──▶  Keyword-based search,
document retrieval
Count / TF-IDF    ──▶   Jaccard Index      ──▶  Tag matching, short text,
deduplication
Word2Vec / GloVe  ──▶   Cosine Similarity  ──▶  Semantic similarity,
paraphrase detection
Word2Vec / GloVe  ──▶   Euclidean Distance ──▶  Clustering similar
documents
Any embedding     ──▶   ML Model (BERT)    ──▶  High-accuracy QA,
chatbots, NLI
```

Scenario	Recommended Pipeline
Fast keyword search	TF-IDF + Cosine Similarity
Short texts / tags	BoW + Jaccard Index
Semantic meaning matters	Word2Vec / GloVe + Cosine Similarity
State-of-the-art accuracy	BERT embeddings + Cosine Similarity
Training data available	Any features + ML classifier

Summary

```
Text #1 ──┐
          ├──▶  Vectorize
──────────────────────────────────────────────┐
Text #2 ──┘      (BoW / TF-IDF / Word2Vec / GloVe / FastText)
│

▼
                                                                Compare ──▶
Score (0–1)
                                                                • Cosine
Similarity  (angle)
                                                                • Euclidean
Distance (distance)
                                                                • Jaccard
Index      (overlap)
                                                                • ML Model
(learned)
```

> **Key insight**: The quality of text similarity depends on how well you vectorize. TF-IDF captures keyword overlap; Word2Vec / GloVe capture meaning. For the best results on real-world tasks, use **dense embeddings + cosine similarity**.

Topic Modeling

Topic modeling is one of the most common applications of NLP.

> We are given a large collection of documents and asked to **"make sense" out of it** — automatically discovering the hidden thematic structure without any labels.

Unlike classification (where you assign a pre-defined label), topic modeling is **unsupervised** — the model discovers topics on its own.

Popular Topic Modeling Algorithms

1. **Latent Dirichlet Allocation (LDA)** — most widely used
2. Latent Semantic Analysis (LSA)
3. Probabilistic Latent Semantic Analysis (PLSA)

Latent Dirichlet Allocation (LDA)

LDA is a probabilistic generative model that assumes:

- Each **document** is a mixture of topics
- Each **topic** is a mixture of words

> "Latent" = hidden (topics are not observed) "Dirichlet" = the probability distribution used to model mixtures "Allocation" = assigning words to topics

Core Assumptions

```
Every document  →  is a blend of K topics  (e.g. 60% Topic A, 40% Topic B)
Every topic     →  is a distribution over words  (e.g. "broccoli" 30%,
"banana" 15%)
```

LDA works **backwards**: given the documents, find the topics and topic-word distributions that most likely generated them.

What Does LDA Do? — Example

Input: 5 Documents

Doc	Text
D1	"I like to eat broccoli and bananas."
D2	"I ate a banana and salad for breakfast."
D3	"Puppies and kittens are cute."
D4	"My sister adopted a kitten yesterday."

Doc	Text
D5	"Look at this cute hamster munching on a piece of broccoli."

LDA is told: *find 2 topics (K = 2)*.

Output: Topics + Document Mixtures

Topic distributions (what words define each topic):

Topic	Top Words	Theme (human-assigned label)
Topic A	broccoli 30%, bananas 15%, breakfast 10%, munching 10%, …	🥦 Food
Topic B	puppies 20%, kittens 20%, cute 20%, hamster 15%, …	🐾 Animals

Document-topic mixtures:

Document	Topic A (Food)	Topic B (Animals)	Dominant Topic
D1 — "broccoli and bananas"	**100%**	0%	Food
D2 — "banana and salad"	**100%**	0%	Food
D3 — "Puppies and kittens"	0%	**100%**	Animals
D4 — "sister adopted a kitten"	0%	**100%**	Animals
D5 — "cute hamster… broccoli"	**60%**	40%	Mostly Food

> LDA does **not** name the topics — it only outputs word probability distributions. Humans look at the top words and assign a label like "Food" or "Animals".

How LDA Works (Intuition)

```
Start:  Randomly assign every word in every document to a topic

Repeat (many iterations):
  For each word w in document d:
    1. Pretend w has no topic assignment
    2. Ask: given the topic mix of d and how common w is in each topic,
            what topic should w belong to?
    3. Reassign w to the most likely topic

After convergence:
  → Topic–word distributions stabilise
  → Document–topic distributions stabilise
```

This iterative process is called **Collapsed Gibbs Sampling** — the standard way to train LDA.

Key Hyperparameters

Parameter	Meaning	Effect
K	Number of topics	Must be chosen before training (try 5–50 for typical corpora)
α (alpha)	Document-topic concentration	Low α → documents focus on few topics; High α → documents spread across many topics
β (beta)	Topic-word concentration	Low β → topics are narrow (specific words); High β → topics are broad

---

LDA vs Other Algorithms

	LDA	LSA	PLSA
Type	Probabilistic (Bayesian)	Linear algebra (SVD)	Probabilistic (MLE)
Handles new docs?	✓ Yes (inference)	✕ Recompute	✕ Recompute
Interpretability	High	Moderate	Moderate
Overfitting risk	Low (Dirichlet prior)	—	High (no prior)
Speed	Moderate	Fast	Moderate
Most popular?	✓ Yes	Common	Rarely used now

Where LDA is Used

Use Case	Example
News organisation	Automatically tag articles by topic (politics, sports, finance)
Customer feedback	Discover complaint themes from thousands of reviews
Research	Find research themes across thousands of academic papers
Recommendation	Suggest articles with overlapping topics
Legal discovery	Group contracts by subject matter

Summary

```
Input:   Collection of documents (no labels)
          │
          ▼
        LDA (K topics)
```

```
            │
            ├──▶  Topic-word distributions
            │     Topic A: {broccoli: 30%, banana: 15%, ...}  → "Food"
            │     Topic B: {puppy: 20%, kitten: 20%, ...}     → "Animals"
            │
            └──▶  Document-topic mixtures
                  D1: 100% Topic A
                  D3: 100% Topic B
                  D5:  60% Topic A, 40% Topic B
```

> **Key insight**: LDA turns an unstructured pile of text into structured, interpretable themes — without any labelled training data. It is the go-to first step for **exploratory analysis** of large text corpora.